

Independent Verification of the European Climate Law's Effect on Nitrogen Dioxide Pollution: A Satellite-Derived, Difference-in-Differences Analysis

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Abstract

Governments frequently claim environmental policy success through self-reported administrative data, yet independent verification using observational evidence remains uncommon in EU climate policy assessment. This study tests whether the European Climate Law (Regulation (EU) 2021/1119, effective 30 June 2021) produced a statistically distinguishable reduction in tropospheric nitrogen dioxide (NO₂) pollution across EU-27 countries, using Sentinel-5P TROPOMI satellite observations (2019–2024) and a Difference-in-Differences (DiD) causal-inference framework. An initial single-cohort model found a statistically significant reduction ($p = 0.026$); however, a placebo test using a counterfactual treatment date revealed this result reflected a general, ongoing European pollution-decline trend rather than a policy-specific effect. Following this finding, a genuine external control group — the United Kingdom, Norway, and Switzerland — was constructed, enabling a properly identified two-group DiD model with country-clustered standard errors. The corrected model found no statistically significant EU-specific effect (coefficient = -1.40×10^{-6} , $p = 0.663$, cluster-robust), independently confirmed via a 23-quarter event-study analysis and a series of additional robustness checks (a bad-control test, a log-transformed outcome, treatment-date sensitivity, and baseline-pollution heterogeneity), none of which overturned the null finding. Applying this same two-group, cluster-robust design to a secondary outcome — NDVI vegetation health, previously assessed only with the original, since-invalidated single-cohort design — revealed a statistically significant relative decline in EU-27 vegetation health versus the control group (coefficient = -0.0210 , $p = 0.012$), reported as an exploratory finding not attributable to the Climate Law itself. These findings suggest that observed EU-wide NO₂ declines over the study period reflect a broader European trend rather than a measurable, policy-specific effect distinguishable at current sample sizes, while the NDVI result illustrates the risk of applying validation rigor unevenly across a study's outcomes. The study's validation methodology — placebo testing, cluster-robust inference, and consistent rigor across primary and secondary outcomes alike — is presented as a methodological contribution as significant as the substantive findings themselves.

Keywords: Difference-in-Differences, satellite remote sensing, air pollution, European Green Deal, causal inference, policy evaluation

1. Introduction

The European Green Deal represents one of the most ambitious environmental policy programs globally, with the European Climate Law establishing a legally binding target of climate neutrality by 2050. Yet the question of whether such legislation produces measurable, attributable environmental improvement — as opposed to reflecting or coinciding with pre-existing trends — is rarely tested with the rigor that causal-inference methodology permits. Existing air quality assessments in Europe often rely on administrative self-reporting or

simple before-after comparisons, both of which are vulnerable to confounding by broader technological, economic, and behavioral trends that would have occurred independent of any specific legislative act.

This study addresses that gap directly: it tests whether the European Climate Law produced a statistically distinguishable reduction in NO₂ pollution — a well-established indicator of combustion-related emissions from transport and industry — using independently observed satellite data rather than administrative claims. This "trust, but verify" approach treats policy effectiveness as a hypothesis to be tested against evidence, not a fact to be assumed.

2. Literature Review

2.1 Satellite-Based Monitoring of NO₂ and Policy Impact

The Sentinel-5P TROPOMI instrument, operated by the European Space Agency since 2017, has become a standard tool for independently observing tropospheric NO₂ concentrations at fine spatial and temporal resolution. A substantial body of research has used TROPOMI data to assess the atmospheric impact of discrete policy events, most extensively during the COVID-19 pandemic. Multiple studies documented double-digit percentage reductions in NO₂ during lockdown periods across European and Asian cities, comparing satellite-observed concentrations against pre-lockdown baselines. Some of this literature has specifically cautioned that simple before-after or year-over-year satellite comparisons risk conflating policy effects with meteorological variability, given the substantial interannual variation in weather conditions relative to the still-limited historical record of the TROPOMI instrument. This concern directly informed the present study's decision to include temperature and precipitation as explicit control variables rather than relying on a raw pollution-level comparison.

Separately, TROPOMI-based work has extended beyond acute lockdown events to structural urban policy: a recent causal-inference evaluation of London's Ultra Low Emission Zone found statistically significant reductions in NO₂ and NO_x following the policy's initial implementation, though not following a subsequent expansion phase, illustrating that satellite-derived causal evaluation can detect genuine, policy-specific effects when methodologically well-identified — reinforcing that a null result under similar methodology is a meaningful finding rather than evidence of the method's insensitivity.

2.2 Difference-in-Differences Methodology and Its Limitations

Difference-in-Differences remains among the most widely used quasi-experimental designs in policy evaluation research, valued for isolating a treatment's effect by comparing outcome changes between a treated and untreated group rather than relying on a single group's before-after comparison alone. Recent methodological literature has emphasized that DiD designs face specific challenges when treatment is not clearly randomized or when the design's core parallel-trends assumption cannot be adequately verified, particularly in settings involving simultaneous, universal policy rollout.

The parallel trends assumption — that treatment and comparison groups would have followed similar trajectories in the absence of intervention — is foundational to DiD's causal validity, yet is frequently difficult to establish empirically, especially over short panels. A widely cited methodological guide recommends placebo regressions, applying the DiD framework to pre-treatment data specifically to confirm that no spurious "treatment effect" is detected before treatment has actually occurred — a diagnostic this study applies directly, with results that proved decisive in revising the analytical approach.

Event-study extensions, which disaggregate an average treatment effect into period-specific estimates, are similarly established practice for testing both pre-trend validity and effect-timing dynamics. Research on pre-treatment significance testing frames event-study coefficients on pre-intervention periods as a direct placebo mechanism, artificially assigning treatment status to periods before the actual intervention to test for spurious differences — the same logic underlying this study's 23-quarter event-study robustness check.

2.3 The Single-Cohort Identification Problem

A methodological challenge specific to EU-wide legislation is that policies frequently apply to all member states simultaneously, leaving no naturally occurring untreated EU comparison group. This is a widely recognized limitation of DiD applications to universal policy rollouts, distinct from settings (such as sub-national or staggered implementation) where genuine within-study controls exist. The present study's initial model encountered this limitation directly, motivating the subsequent introduction of a non-EU comparison group — a design choice consistent with established practice of seeking geographically and structurally comparable jurisdictions outside a policy's legal jurisdiction when no internal control exists.

3. Data and Methodology

3.1 Study Design

This study evaluates the causal effect of the European Climate Law (effective 30 June 2021) on tropospheric NO₂ concentration using a panel Difference-in-Differences design across 30 European countries: the EU-27 (treatment group) and three non-EU comparator countries — the United Kingdom, Norway, and Switzerland (control group) — selected for geographic proximity, economic development comparability, and explicit non-applicability of EU Green Deal legislation.

Because the Climate Law applied to all EU-27 member states simultaneously on a single effective date, this design has a single treatment cohort and a single treatment timing, with no staggered rollout across units. This structurally avoids the negative-weighting and heterogeneous-treatment-effect bias that recent econometric literature has identified in two-way fixed-effects DiD estimators specifically when treatment timing varies across units (Callaway, Goodman-Bacon & Sant'Anna, 2024) — a critique this study's design is not exposed to, though the absence of any staggered timing is precisely what removed the possibility of a naturally occurring within-EU control group in the first place (Section 2.3).

GPIE Study Design: Treatment vs. Control Group
Difference-in-Differences comparison — EU-27 (subject to the European Climate Law) vs. three non-EU comparator countries

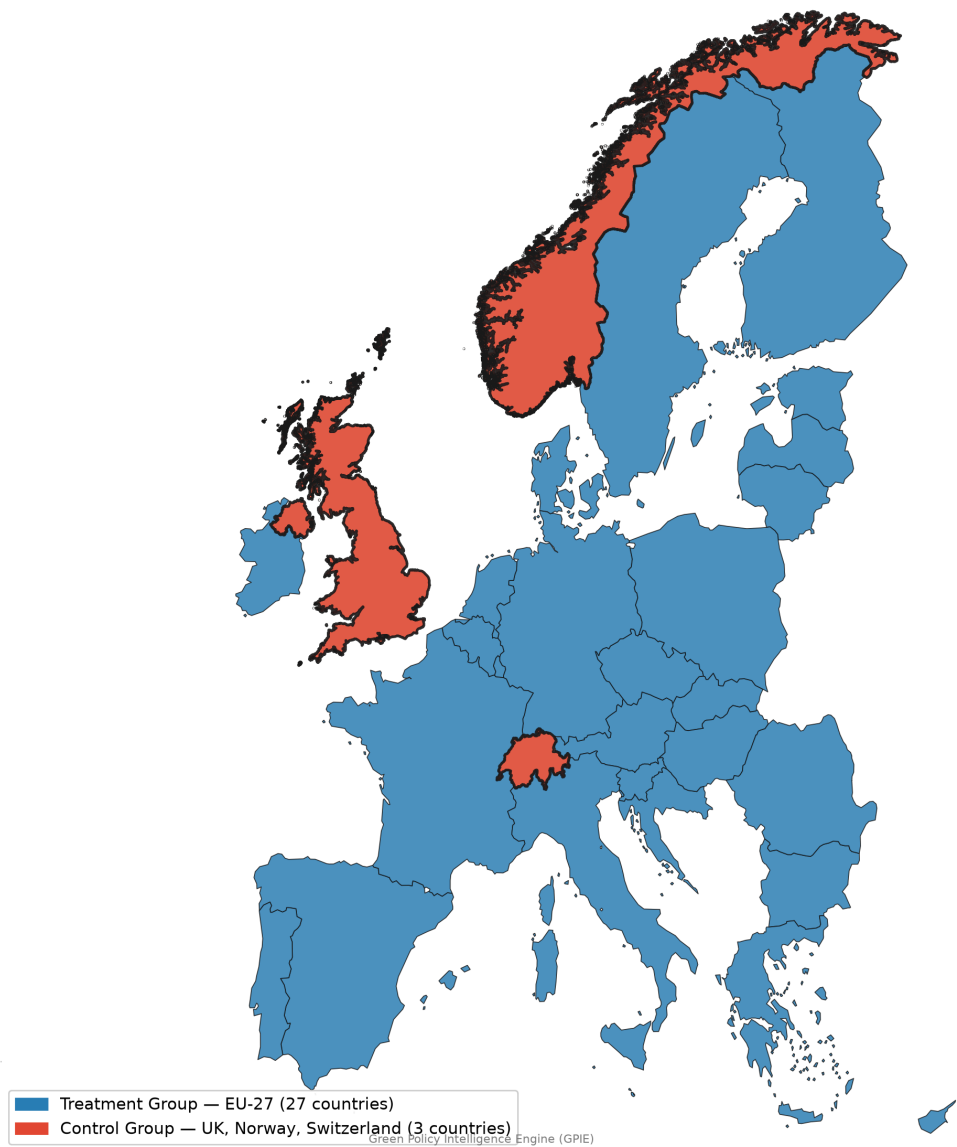


Figure 1. Difference-in-Differences study design showing the treatment group (EU-27) and the non-EU control group (United Kingdom, Norway, and Switzerland).

3.2 Data Sources

Variable	Source	Resolution
NO ₂ (outcome)	Sentinel-5P TROPOMI, via Sentinel Hub Statistical API	Monthly, country-level
NDVI (secondary outcome)	Copernicus Global Land Service	Monthly, country-level
Temperature, precipitation (controls)	ERA5 Reanalysis, Copernicus Climate Data Store	Monthly, country-level
GDP (control)	Eurostat (EU-27); World Bank (control group)	Annual, country-level

Variable	Source	Resolution
Administrative boundaries	Eurostat GISCO (NUTS); GADM (control group)	—

The panel spans January 2019 to December 2024 (72 months), providing 30 months of pre-treatment and 42 months of post-treatment observation.

3.3 Model Specification

The final model estimates:

$$NO2_{it} = \beta_0 + \beta_1(Treatment_i \times Post_t) + \beta_2Post_t + \gamma X_{it} + \alpha_i + \delta_m + \varepsilon_{it}$$

where $Treatment_i$ indicates EU-27 membership, $Post_t$ indicates observations after 30 June 2021, X_{it} is a vector of time-varying controls (temperature, precipitation, GDP), α_i are country fixed effects, and δ_m are calendar-month fixed effects controlling for seasonality. The coefficient of interest, β_1 , captures the EU-specific effect of the Climate Law, net of any trend common to both treatment and control groups. All standard errors are clustered by country, since panel data with repeated monthly observations per country exhibits within-country serial correlation that uncorrected (classical) standard errors do not account for, understating true uncertainty (Bertrand, Duflo & Mullainathan, 2004).

3.4 Robustness Checks

Two robustness procedures were applied prior to accepting any result: a **placebo test**, re-estimating the model with an artificial treatment date (30 June 2020) at which no comparable policy event occurred, and an **event-study specification**, replacing the single post-treatment indicator with 23 quarter-specific interaction terms (2019Q1–2024Q4, relative to a 2021Q2 reference period) to test both pre-treatment parallel trends and potential delayed treatment effects.

4. Results

4.1 Initial Single-Cohort Model

A first model, comparing all 27 EU countries before and after treatment without an external control group, found a statistically significant NO₂ reduction (coefficient = -2.29×10^{-6} , p = 0.026, 95% CI [-4.30×10^{-6} , -2.69×10^{-7}]).

4.2 Placebo Test

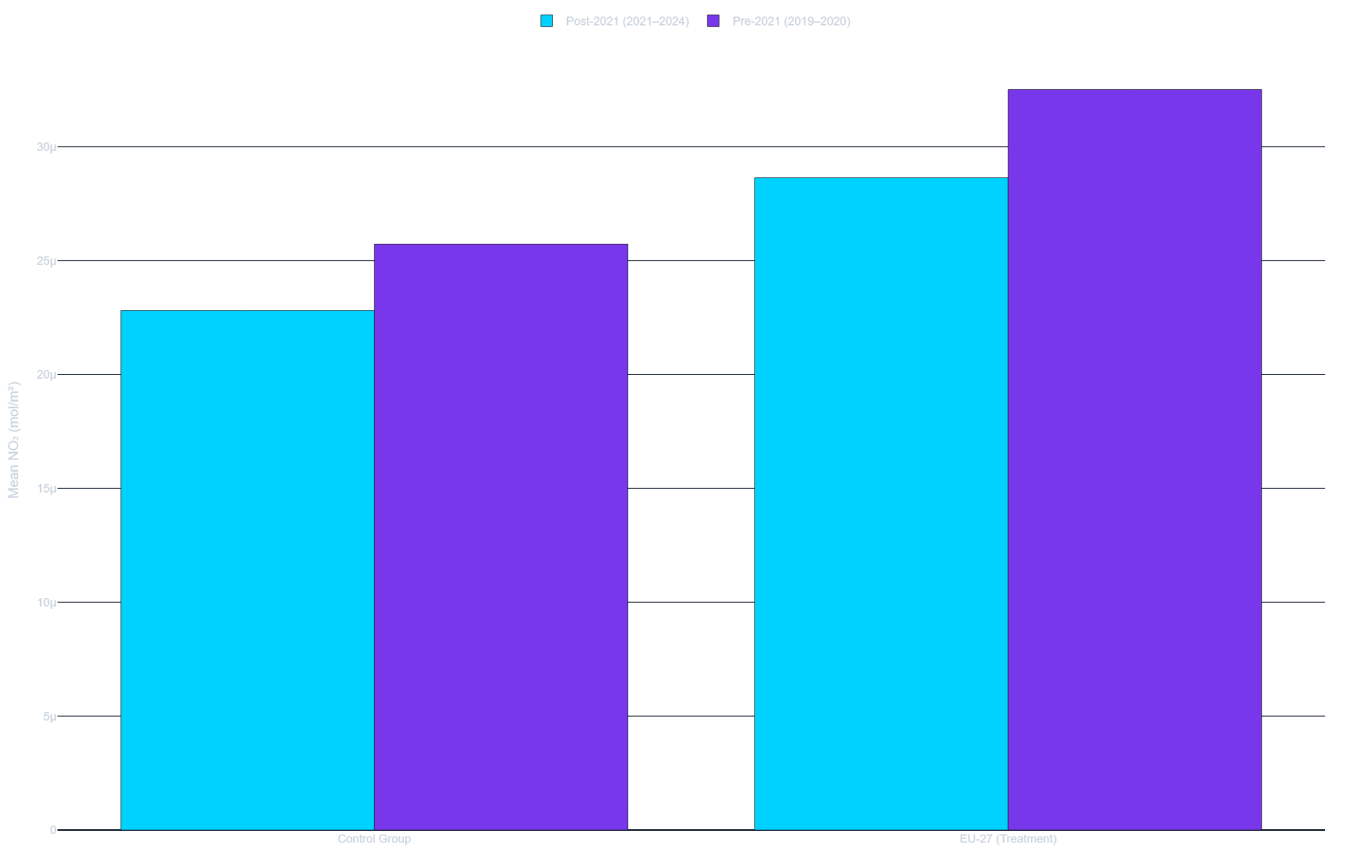
Re-estimating the identical model with the treatment date shifted to 30 June 2020 produced a coefficient of -3.29×10^{-6} with p = 0.004 (cluster-robust standard errors, clustered by country) — a result more statistically significant than the genuine treatment date. This outcome is inconsistent with a correctly identified policy effect and indicates the original model was capturing a general, ongoing pollution-decline trend rather than an effect specific to the Climate Law. Adding an explicit linear time trend to the original model confirmed this: the coefficient at the true treatment date became statistically indistinguishable from zero (p = 0.186, cluster-robust) once the underlying trend was controlled for directly.

4.3 Corrected Two-Group Model

Following construction of the non-EU control group, the corrected DiD model produced:

Statistic	Value
DiD coefficient	-1.40×10^{-6}
P-value (cluster-robust, by country)	0.663
95% Confidence Interval	$[-7.68 \times 10^{-6}, +4.88 \times 10^{-6}]$
R ²	0.386
N	1,930

Figure 2. Average NO₂ concentrations before (2019–2020) and after (2021–2024) the European Climate Law for the EU-27 treatment group and the non-EU control group. Both groups exhibit a similar decline, visually supporting the Difference-in-Differences estimate that no statistically distinguishable EU-specific treatment effect was detected.



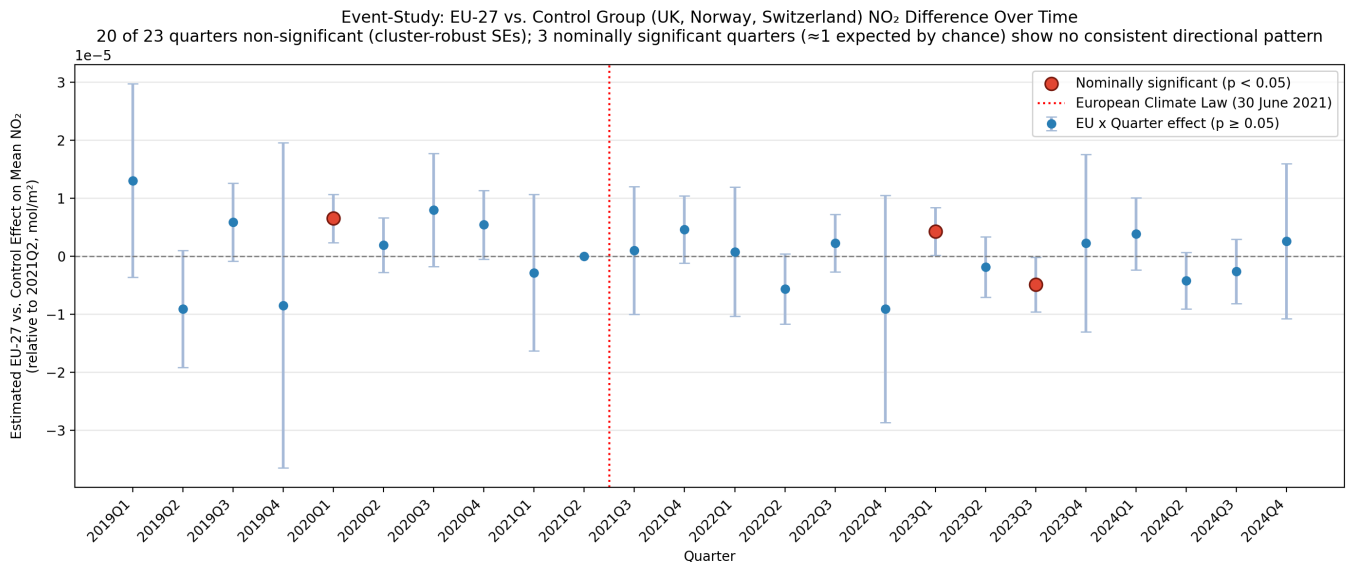
The interaction term is not statistically significant, and its confidence interval spans zero.

4.4 Event-Study Validation

Under cluster-robust standard errors, 20 of the 23 quarterly interaction coefficients remain non-significant ($p > 0.05$). Three quarters are nominally significant: 2020Q1 (pre-treatment, coefficient positive), and 2023Q1 and 2023Q3 (post-treatment, opposite signs). This is close to the ~ 1 false positive expected by chance across 23 independent tests at the 5% level, and the three flagged quarters do not form a consistent directional pattern — the pre-treatment case (2020Q1) plausibly reflects the onset of COVID-19 lockdowns, which affected EU and non-EU European countries on different timelines, rather than a genuine pre-existing divergence between the treatment and control groups; the two post-treatment quarters point in opposite directions and are separated by a year, inconsistent with an emerging or delayed treatment effect. Taken together, the event-

study results support the parallel-trends assumption and do not indicate a delayed effect emerging at any point through the end of 2024.

Figure 3. Event-study estimates showing quarter-specific Difference-in-Differences coefficients relative to the 2021Q2 reference period. Error bars represent 95% confidence intervals. No quarter exhibits a statistically significant deviation from zero before or after the European Climate Law.

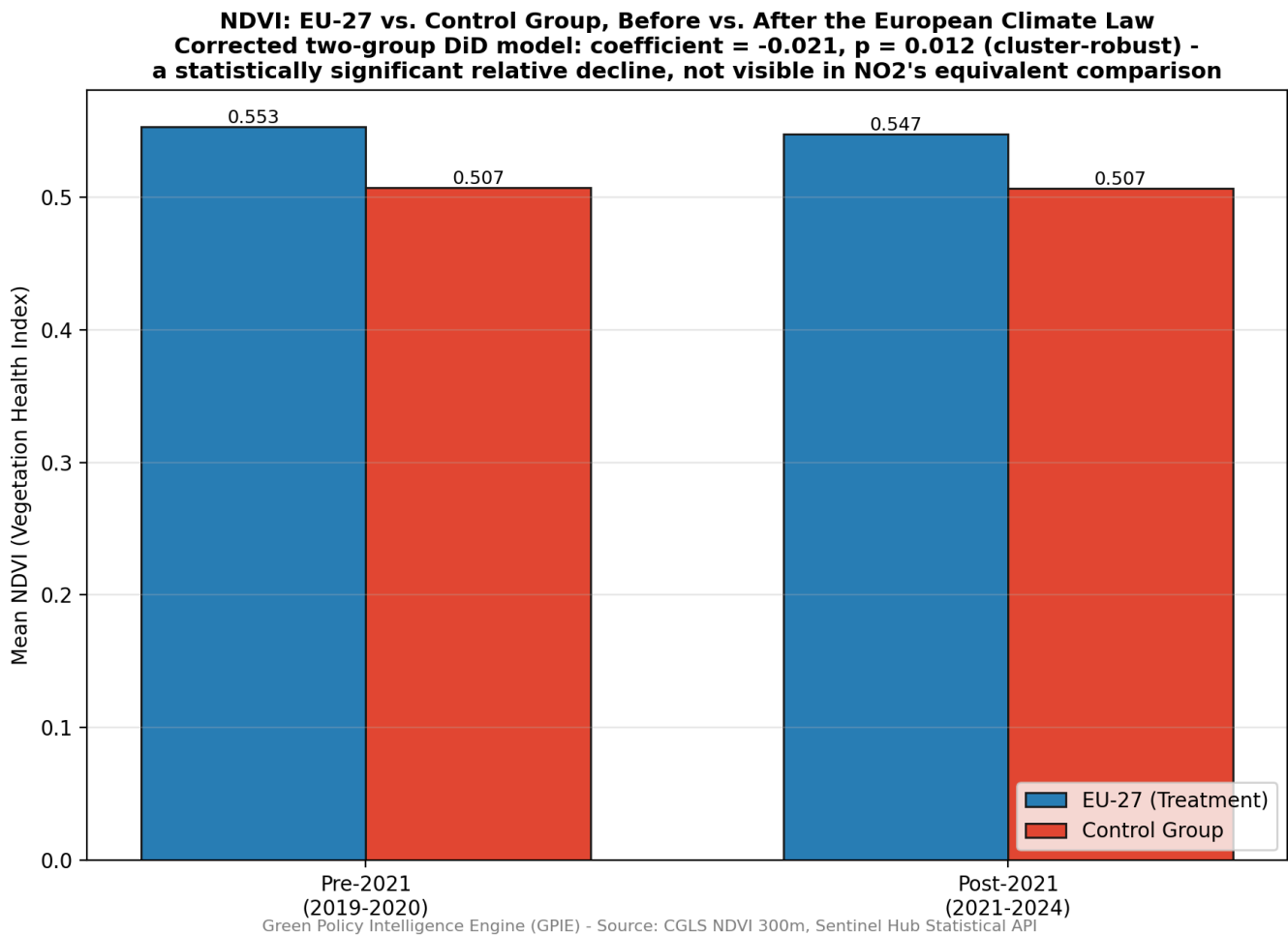


4.5 Secondary Outcome (NDVI)

An initial NDVI model, mirroring the study's original single-cohort NO₂ specification (EU-27 only, no external control group), found no significant effect (coefficient = -0.0059 , $p = 0.128$). However, since the primary NO₂ analysis had already demonstrated via the placebo test (Section 4.2) that a single-cohort design cannot reliably distinguish a policy-specific effect from a general regional trend, the same two-group control-group correction applied to NO₂ was subsequently applied to NDVI as well, using the identical treatment-group $\times$ post-treatment specification and the genuine non-EU control group.

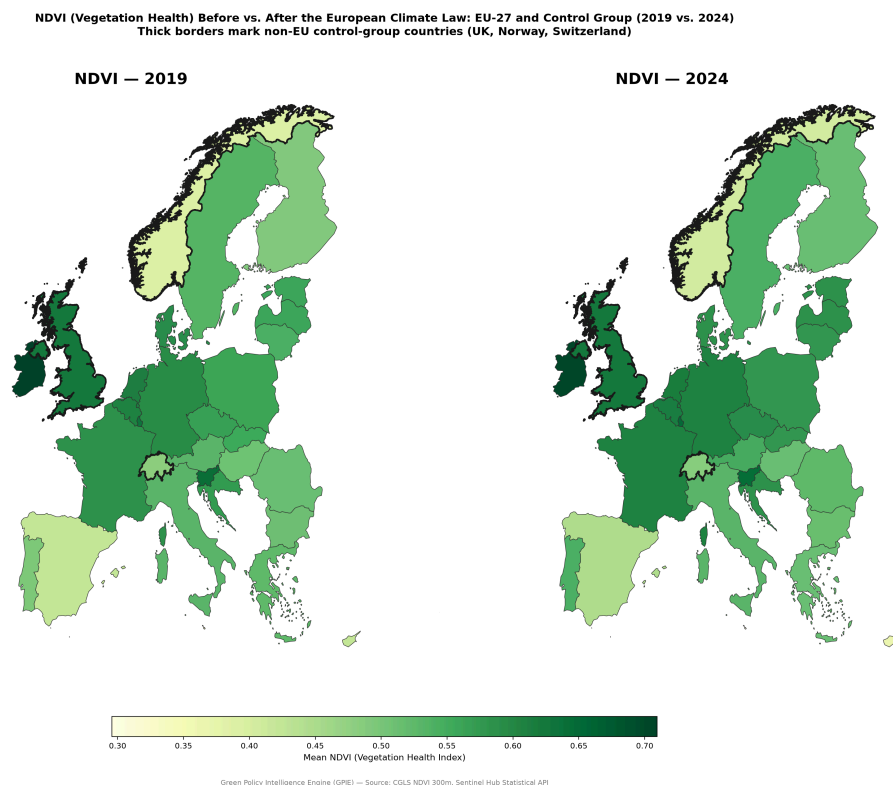
This corrected two-group model produced a materially different result: coefficient = -0.0210 , $p = 0.012$ (cluster-robust standard errors, by country), 95% CI [-0.0372 , -0.0047] — a statistically significant *decline* in EU-27 NDVI relative to the non-EU control group following the Climate Law's effective date. This is a genuinely different finding from the originally reported null result, uncovered specifically by applying the same methodological rigor already used for the primary outcome.

Figure 4. Mean NDVI for the EU-27 treatment group and the non-EU control group, before (2019–2020) and after (2021–2024) the European Climate Law. Unlike the equivalent NO₂ comparison (Figure 2), the two groups' NDVI trajectories diverge — the visual basis for the significant relative decline reported above.



This result should be interpreted cautiously rather than read as evidence that the Climate Law itself reduced vegetation health. The Climate Law is an emissions-focused instrument, not a land-use or vegetation-management policy, and this analysis does not control for land-use change, drought or precipitation-driven vegetation stress differences between the treatment and control regions, or shifts in agricultural policy over the same period — any of which could plausibly drive a relative NDVI decline unconnected to the Climate Law specifically. The static, single-snapshot land-cover control included in this study's broader design (Section 3.2) cannot capture such time-varying confounds. This finding is reported as an honest, statistically supported secondary result meriting further investigation — with land-use and agricultural-policy controls as a natural next step — rather than as a causal claim about the Climate Law's effect on vegetation.

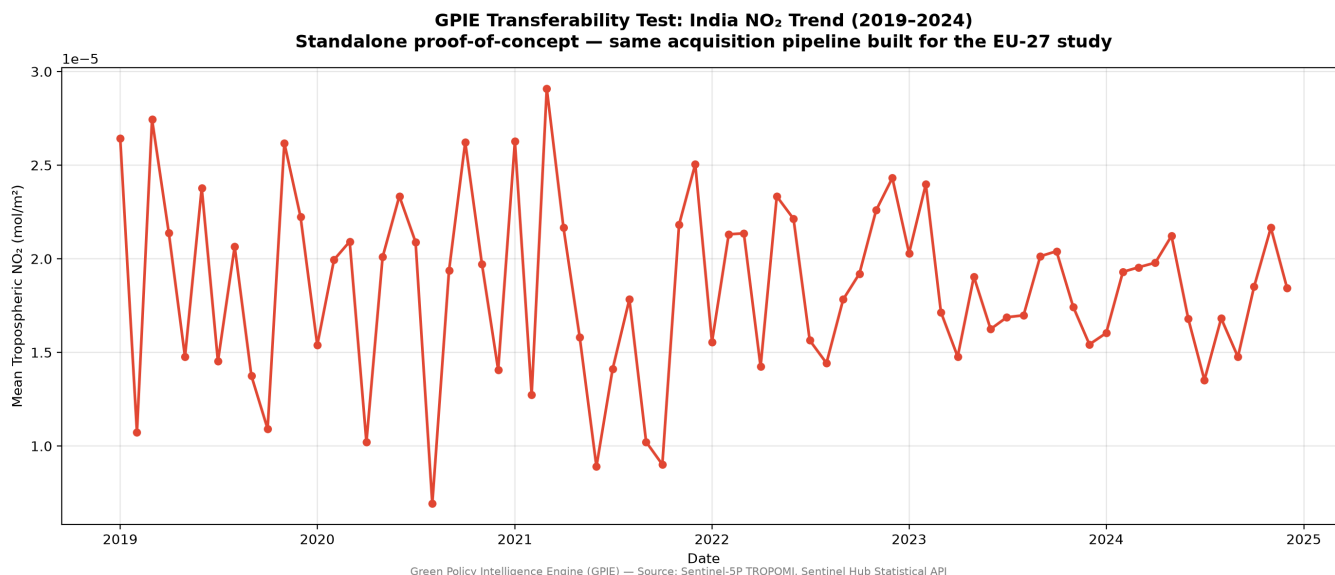
Figure 5. Average NDVI before (2019–2020) and after (2021–2024) across the EU-27 study area. The corrected two-group Difference-in-Differences estimate (Section 4.5) finds a statistically significant relative decline versus the non-EU control group, though the magnitude is subtle enough that it is not necessarily obvious from visual inspection of EU-27 values alone.



4.6 Transferability Validation

To provide direct evidence for this framework's applicability beyond the EU-27 study region — rather than presenting transferability as an unverified design claim — the NO_2 acquisition pipeline was independently tested on India (2019–2024), using identical Sentinel Hub Statistical API infrastructure with no modification to the core acquisition logic. All six years were retrieved successfully, returning physically realistic NO_2 concentrations consistent with the range observed across the EU-27 dataset. This is presented as a standalone architectural validation, not a comparative causal analysis; establishing India as a genuine study case would require the same rigor (control-group construction, placebo testing) applied to the EU-27 analysis in this paper.

Figure 6. Transferability validation using India as an independent test case. The identical Sentinel Hub acquisition pipeline successfully retrieved physically realistic NO_2 observations for 2019–2024 without modification, demonstrating that the framework is geographically transferable beyond the original EU-27 study area.



4.7 Additional Robustness Checks

GDP as a potential "bad control." GDP is a time-varying covariate that could itself be affected by climate policy (or serve as a channel through which policy affects emissions), which risks biasing a DiD estimate if included naively (Angrist & Pischke, 2009). Re-estimating the corrected two-group model with GDP removed entirely produces a coefficient of -4.80×10^{-7} ($p = 0.880$, cluster-robust) — closer to zero and less significant than the model including GDP, not more. This indicates GDP is not driving, inflating, or masking the reported null result; if anything, its inclusion makes the estimate more conservative rather than less. A specification using only each country's pre-treatment average GDP (rather than the time-varying series) is numerically identical to the no-GDP specification, since a time-invariant country-level covariate is fully absorbed by the model's country fixed effects.

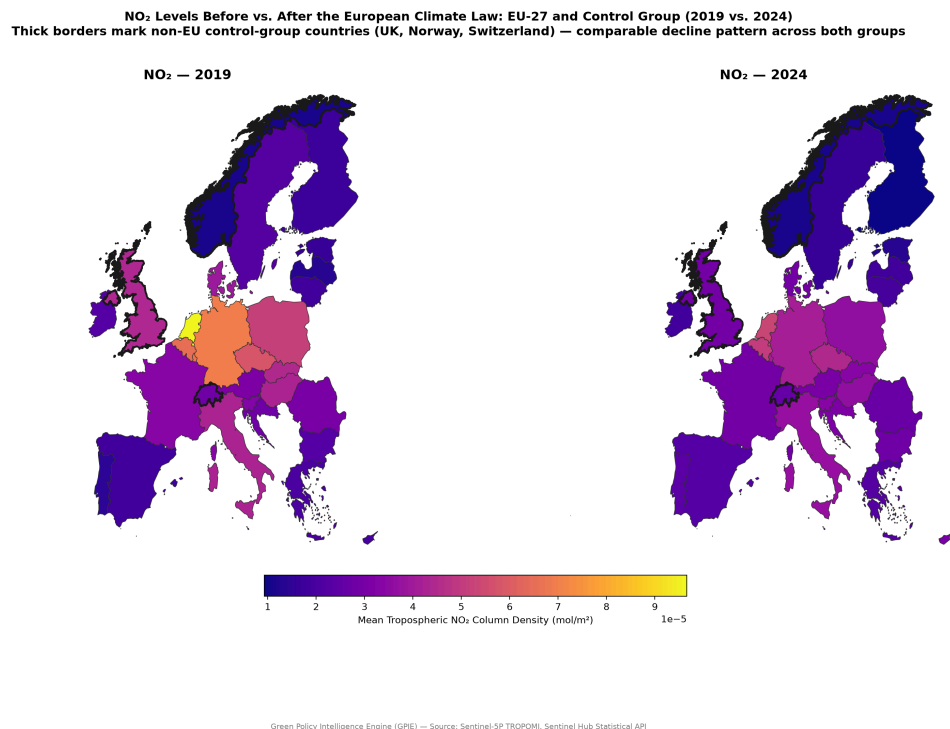
Functional form: log-transformed outcome. Since pollution concentrations are typically right-skewed, a log-transformed specification was estimated as a robustness check to the linear-level model. Twenty-three of 1,930 country-month observations (1.2%) have non-positive mean NO₂ values, a known characteristic of satellite-derived trace-gas retrievals at very low concentrations, where measurement noise can produce small negative readings around a near-zero true value; ten of these twenty-three are concentrated in December 2023 across multiple countries, consistent with a previously identified data-acquisition gap for that month (documented in the project's development log) rather than twenty-three independent anomalies. Excluding these observations, the log-transformed model finds a coefficient of 0.046 (approximately a 4.7% relative change, $p = 0.669$, cluster-robust), consistent with the linear model's null result and confirming the finding is not an artifact of functional-form choice.

Treatment-date sensitivity. Since the treatment date is anchored to a single legal instrument rather than a graduated policy-intensity measure (Section 6), the model was re-estimated with the assumed treatment date shifted by ± 6 and ± 12 months (30 June 2020, 31 December 2020, 31 December 2021, and 30 June 2022). No shifted date produced a statistically significant effect at the 5% level ($p = 0.764, 0.357, 0.151$, and 0.086 respectively, versus $p = 0.663$ at the true date), indicating the null result is not an artifact of one specific date choice. The +12-month date is the closest to conventional significance ($p = 0.086$); however, the event-study's more granular quarter-by-quarter disaggregation (Section 4.4) found no consistent post-treatment pattern around that period, suggesting this is consistent with the wider sampling variability inherent in a two-period split rather than a genuine delayed effect.

Heterogeneity by baseline pollution level. The pooled EU-27 estimate could mask an effect concentrated in a subset of countries. The treatment group was split at the median pre-treatment NO₂ level into thirteen higher-baseline-pollution countries (largely Western/Central European) and fourteen lower-baseline-pollution countries (largely Southern/Eastern European and smaller economies), each re-estimated separately against the full non-EU control group. Neither subgroup shows a statistically significant effect (higher-baseline: coefficient = -4.49×10^{-6} , $p = 0.245$; lower-baseline: coefficient = $+3.74 \times 10^{-6}$, $p = 0.339$), though the point estimates diverge in direction. Given that splitting the sample roughly halves the statistical power available to the pooled model, this divergence is not strong evidence of a genuine subgroup effect, but it is flagged as a candidate direction for future work with a larger or extended panel.

5. Discussion

Figure 7. Average tropospheric NO₂ concentrations across the EU-27 in 2019 (pre-treatment) and 2024 (post-treatment). Although a visual reduction is evident, visual change alone cannot establish causality. The Difference-in-Differences analysis demonstrates that a similar decline occurred in the non-EU control group, indicating the observed reduction reflects a broader European trend rather than a statistically distinguishable effect of the European Climate Law.



The corrected model's finding — no statistically distinguishable EU-specific NO₂ reduction attributable to the Climate Law — should be interpreted against the demonstrated capacity of this same broad methodological family to detect genuine policy effects: comparable satellite-based DiD analyses of sub-national policy interventions, such as London's low-emission zone, have found clear, significant effects. This makes it more plausible that the present null result reflects a genuine absence of a detectable EU-specific effect, rather than a general insensitivity of satellite-based DiD methodology to real policy impacts.

The result is also consistent with the possibility that observed European NO₂ decline over the study period is substantially driven by factors common to both EU and non-EU countries alike — vehicle fleet modernization, broader decarbonization of electricity generation, and pre-existing regulatory momentum predating the Climate Law specifically — none of which would be captured as an EU-specific effect in this design.

A further consideration is whether the non-EU control group satisfies the stable-unit-treatment-value assumption (SUTVA) that DiD requires — specifically, whether the control units are genuinely unaffected by the treatment. NO₂ disperses physically across borders, and the United Kingdom, Norway, and Switzerland are geographically adjacent to and economically integrated with the EU-27; if the Climate Law had a real effect, some of it could plausibly transmit into neighboring control-country readings through atmospheric transport or through EU-linked trade and industrial activity. This would not invalidate the null finding as reported, but it would work in a specific, identifiable direction: any such spillover would attenuate the estimated treatment effect toward zero, meaning this study's null result is, if anything, a conservative one, and cannot be interpreted as ruling out a real effect that leaks into the control group by design. Testing for this directly would require sub-national, border-proximate monitoring data that this country-level analysis does not include, and is noted here as a specific, well-defined direction for future refinement rather than a flaw this study's data could resolve.

The significant relative decline found in the secondary NDVI analysis (Section 4.5) presents a more complex picture than the primary NO₂ result. Unlike NO₂, where the corrected model confirms the initial finding was spurious, NDVI shows the opposite pattern: an initially null single-cohort result becomes significant once the same control-group correction is applied. Given that the European Climate Law is not primarily a land-use or vegetation policy, this study does not interpret the finding as evidence that the Climate Law reduced vegetation health, but rather flags it as a genuine, methodologically robust anomaly that current controls cannot explain — and one that would not have been detected without applying the primary outcome's own validation standard to the secondary outcome as well.

6. Limitations

Several limitations should be considered when interpreting these findings. First, the control group comprises only three countries, a comparatively small comparison group for a panel model with extensive fixed-effects structure; the resulting confidence interval is wide rather than tightly bounded around zero, meaning the null result is consistent with either a genuinely negligible effect or insufficient statistical power to detect a real but modest one. Quantifying this directly: at 80% power and $\alpha = 0.05$, this design's minimum detectable effect is approximately 28.4% of the EU-27's pre-treatment average NO₂ level — this study can rule out an effect of that size or larger with reasonable confidence, but cannot rule out a smaller true effect. The observed coefficient itself corresponds to roughly 4.4% of the pre-treatment baseline, well below this design's detection threshold; the null result should therefore be read as "no effect of at least ~28% detected," not as evidence against smaller effects. Second, the treatment date was anchored to a single, most legally significant instrument (the European Climate Law) rather than a comprehensive measure of cumulative Green Deal policy intensity, since available policy-database records did not support a more granular, country-differentiated treatment measure. Third, currency conversion for the control group's GDP data relied on approximate annual exchange rates rather than precise historical rates, an acceptable approximation for a control variable but not for a variable of primary interest. Fourth, WorldPop population data was available only for 2019–2020 within the accessible dataset version at the time of acquisition; population was therefore treated as a supporting, descriptive variable rather than a model input, given its incomplete temporal coverage relative to the study period. Fifth, the significant NDVI finding (Section 4.5) is not controlled for time-varying land-use change, drought/precipitation-driven vegetation stress, or agricultural policy shifts between the treatment and control regions; the study's land-cover control is a static, single-snapshot variable and cannot capture such dynamics, so this result should be treated as an exploratory secondary finding rather than a fully identified causal estimate.

An originally planned module ranking Green Deal policies by cost-effectiveness (cost per unit of environmental improvement) was deliberately not pursued once Section 4 established that no statistically significant effect had been detected: ranking interventions against an effect size indistinguishable from zero would require either manufacturing significance the data does not support, or ranking against noise, neither of which is scientifically defensible. This module was scoped out on principle rather than left incomplete, consistent with this study's broader commitment to letting the validated result — rather than a downstream deliverable's requirements — determine what analysis is appropriate to report.

7. Conclusion

This study finds no statistically distinguishable, EU-specific reduction in NO₂ pollution attributable to the European Climate Law, once rigorously tested against a genuine non-EU control group and validated through placebo testing and event-study disaggregation. This null result followed directly from a validation process that identified and corrected a flawed initial specification — itself demonstrating the practical necessity of placebo testing in policy evaluation research applying single-cohort designs to universally applied legislation. Applying that same validation standard to the secondary NDVI outcome, rather than stopping at the primary result, surfaced a significant relative decline that the original single-cohort NDVI specification had missed entirely — underscoring that methodological rigor applied unevenly across primary and secondary outcomes can itself become a source of error. The framework developed here is directly transferable to future evaluations of EU environmental policy, and to other jurisdictions facing the same structural identification challenge of legislation without a naturally occurring internal comparison group — a claim supported empirically in this study by a successful standalone acquisition test on India (Section 4.6), rather than asserted without evidence.

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Full dataset, code, and reproducible pipeline: github.com/sakshimaske303-commits/GPIE **Live interactive dashboard:** f5cf6fijj9gm564r6aapt6.streamlit.app